LLM Hallucination Detection via Uncertainty

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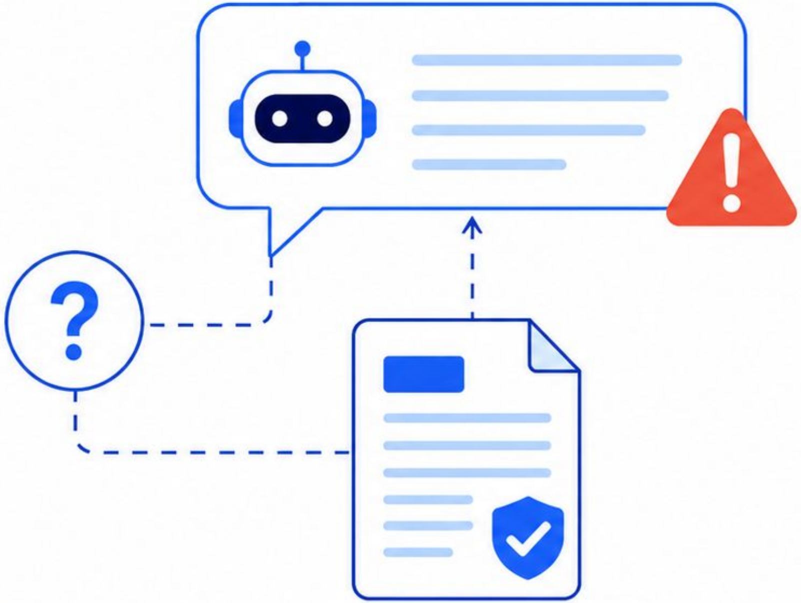
SEDS537 Machine Learning

Problem Motivation

- LLMs can generate fluent but factually wrong answers
- These errors are called hallucinations
- Hallucinations reduce trust in factual QA systems
- The goal is to detect unsupported or incorrect answers automatically



Project Approach

- Qwen is used as a feature extractor, not a judge
- Candidate answers are scored token by token
- Probability, log-probability, and entropy are extracted
- Classical ML classifiers predict the final label

Datasets

HaluEval QA

- Context-grounded QA
- Supported and hallucinated answers
- 20,000 processed examples
- Also tested without context

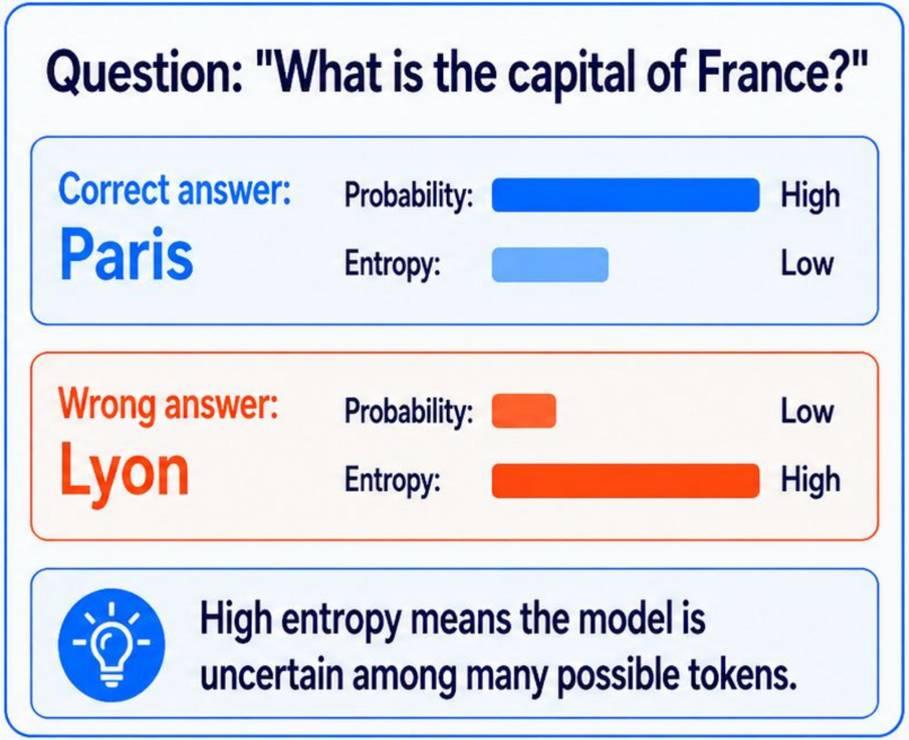
TruthfulQA

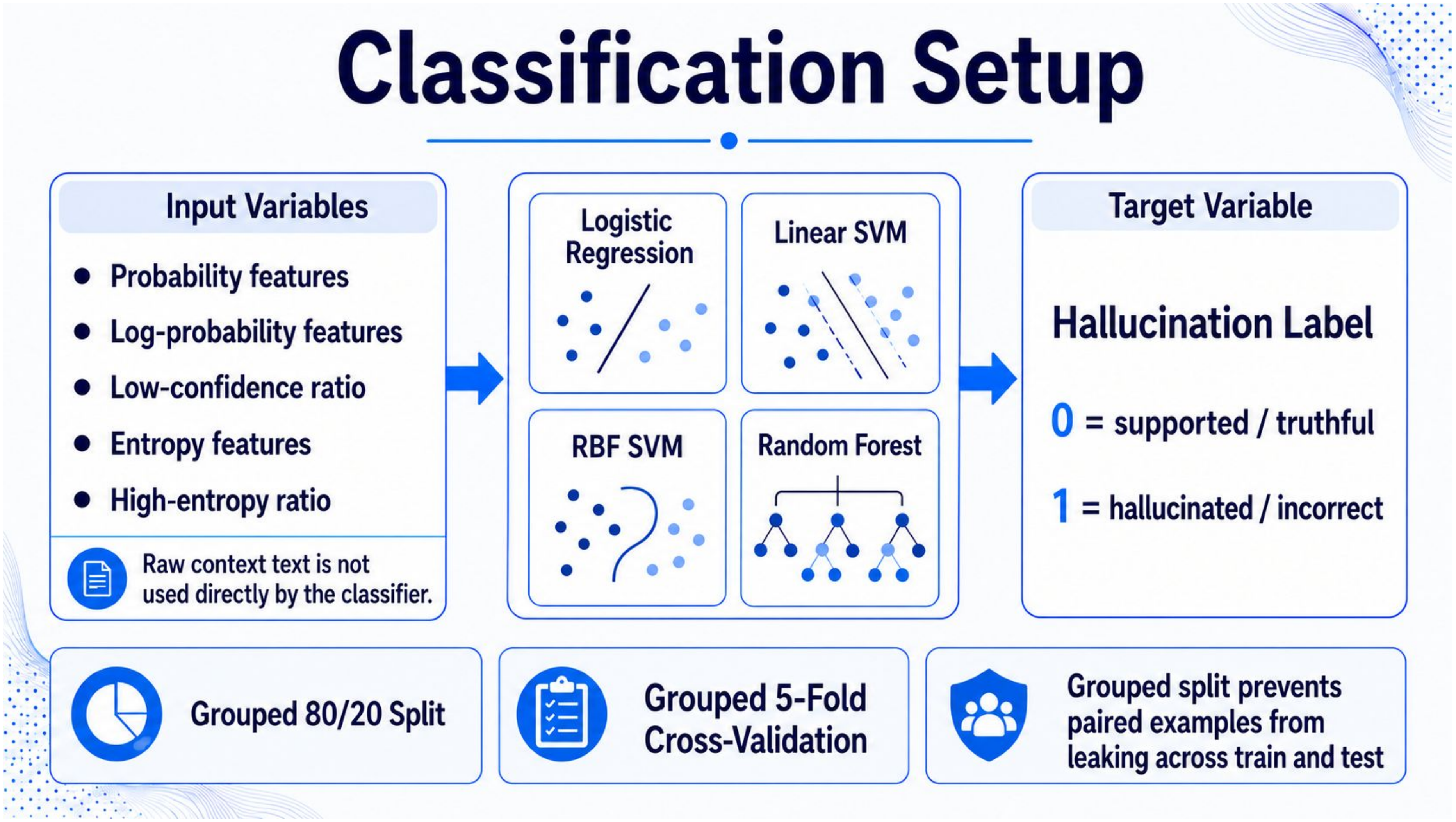
- No-context truthfulness dataset
- Focuses on misconceptions
- 5,918 processed examples
- Harder generalization test

Label 0 = supported/truthful | Label 1 = hallucinated/incorrect

Uncertainty Features

- Probability features: measure how confident the model is about the answer tokens.
- Log-probability features: measure the likelihood of the answer in a more stable mathematical form.
- Low-confidence ratio: measures what fraction of answer tokens received low probability.
- Entropy features: measure how uncertain the model was among possible next tokens.
- High-entropy ratio: measures what fraction of answer positions had high uncertainty.




Classification Setup
Input Variables
Probability features
Log-probability features
Low-confidence ratio
Entropy features
High-entropy ratio
Raw context text is not used directly by the classifier.
Logistic Regression
Linear SVM
RBF SVM
Random Forest
Target Variable
Hallucination Label
0 = supported / truthful
1 = hallucinated / incorrect
Grouped 80/20 Split
Grouped 5-Fold Cross-Validation
Grouped split prevents paired examples from leaking across train and test

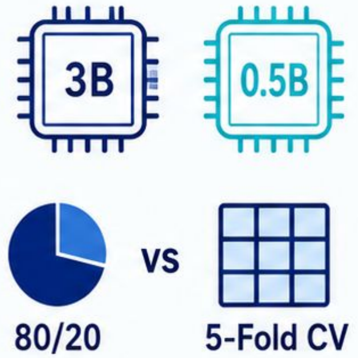
Main Results: Model Size and Split Comparison

Best F1 Scores by Dataset, Split Type, Feature Extractor, and Classifier

Dataset / Setting	Split	Best Model	Qwen2.5-3B F1	Qwen2.5-0.5B F1
HaluEval QA	80/20	Linear SVM / RBF SVM	0.9885	0.9762
HaluEval QA	5-Fold CV	RBF SVM	0.9883	0.9759
HaluEval no-context	80/20	Random Forest	0.9276	0.9283
HaluEval no-context	5-Fold CV	Random Forest	0.9277	0.9271
TruthfulQA	80/20	RBF SVM	0.7037	0.7150
TruthfulQA	5-Fold CV	RBF SVM	0.7173	0.7181

Best Model is selected by highest F1.

For HaluEval QA 80/20, the best model differs by feature extractor.



Dataset difficulty matters more than model size or split type.

Effect of Context

- HaluEval with context gives the strongest result
- Removing context lowers F1 from 0.9883 to 0.9277
- Context helps Qwen assign more useful uncertainty scores
- But HaluEval no-context is still easier than TruthfulQA
- This shows that dataset structure also matters

With Context:
F1 0.9883

No Context:
F1 0.9277

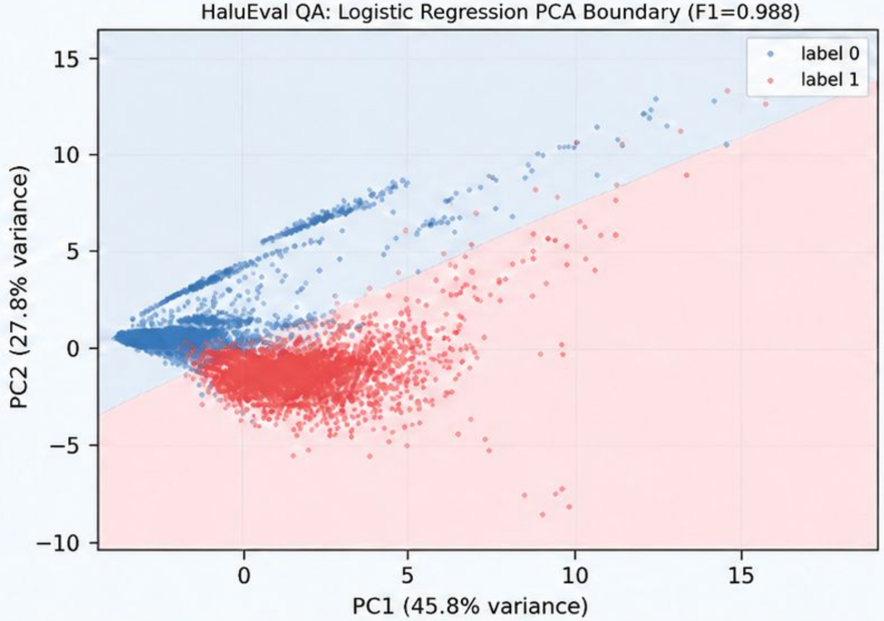
TruthfulQA:
F1 0.7173

Visual Analysis: PCA Decision Boundaries

HaluEval QA

F1 ≈ 0.99

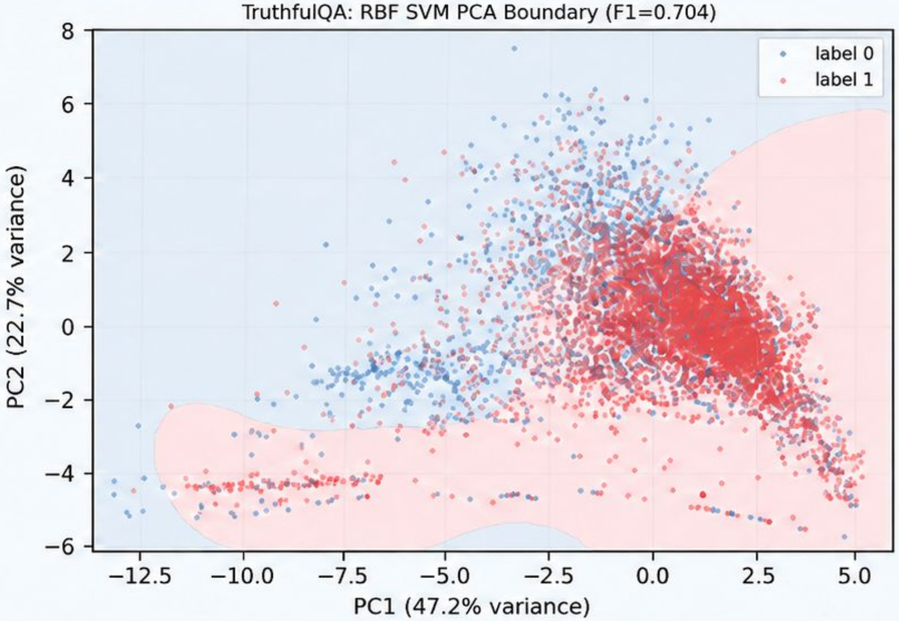
- Clearer class separation
- Easier classification



TruthfulQA

F1 ≈ 0.70

- More class overlap
- Harder classification



TruthfulQA is harder because uncertainty features are less separable.

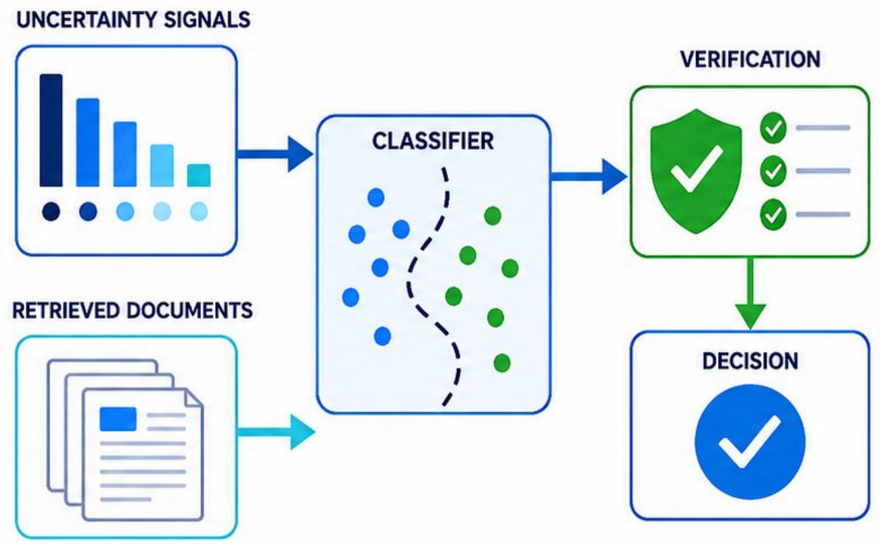
Ablation and Error Analysis

- Full feature set gives the best HaluEval result
- Entropy is especially useful on TruthfulQA
- TruthfulQA false answers can sound familiar to the model
- Truthful answers may reject common myths and look uncertain
- Therefore, confidence is not always the same as truth

Feature Group	HaluEval F1	TruthfulQA F1
Confidence + Logprob	0.9858	0.6904
Entropy	0.9720	0.7160
All Features	0.9885	0.7037

Conclusion

- Uncertainty features are effective for context-grounded hallucination detection
- HaluEval QA achieved very high F1 scores
- TruthfulQA remains difficult because it tests misconceptions
- Uncertainty alone is not enough for robust no-context truthfulness detection
- Future work should add retrieval, source verification, and self-consistency



Uncertainty is useful, but evidence is still needed.